MEMBERS of class	visibility modes –protected for acquiring the properties of the base class		
	Private	Protected	Public
Shape(base class)	int count;	int width; int height;	void setWidth(int) void setHeight(int)
Rectangle (derived class acquired the properties of base class with protected visibility)	Private members of base classes are not directly accessible by the derived class	int width; int height; void setWidth(int) void setHeight(int)	int getArea();

In case the class rectangle is derived with private visibility mode from its base class shape then the property of class rectangle will change as follows

MEMBERS of class	visibility modes –private for acquiring the properties of the base class		
	Private	Protected	Public
Shape(base class)	int count;		void setWidth(int) void setHeight(int)
Rectangle (derived class acquired the properties of base class with private visibility)	int width; int height; void setWidth(int) void setHeight(int)		int getArea();

When you derive a class from an existing base class,it may inherit the properties of the base class based on its visibility mode.So one must give appropriate visibility mode depending on the need.

Private inheritance should be used when you want the features of the base class to be available to the derived class but not to the classes that are derived from the derived class.

Protected inheritance should be used when features of base class to be available only to the derived class members but not to the outside world.

Public inheritance can be used when features of base class to be available to the derived class members and also to the outside world.

16.6 Inheritance and constructors and destructors

When an object of the derived class is created ,the compiler first call the base class constructor and then the constructor of the derived class. This is because the derived class is built up on the members of the base class. When the object of a derived class expires first the derived class destructor is invoked followed by the base class destructor.

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